

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 03

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 Evaluate $\binom{10}{3} - \binom{10}{7} + \binom{9}{2}$.

A 36

B 240

C 0

D 84

E 120

F 156

2 Find the sum to infinity of the geometric series $12 - 4 + \frac{4}{3} - \frac{4}{9} + \dots$

A 16

B 6

C 9

D 18

E 24

F 8

3 Simplify $\frac{1 - \cos^2 \theta}{\cos \theta \tan \theta}$ for values of θ where the expression is defined.

A 1

B $\sin \theta$

C $\tan \theta$

D $\sin^2 \theta$

E $\cos \theta$

F $\frac{\sin \theta}{\cos \theta}$

4 Claim: if $a > b$ then $\frac{1}{a} < \frac{1}{b}$.

I Suppose $a > b$.

II Dividing both sides by ab : $\frac{1}{b} > \frac{1}{a}$.

III So $\frac{1}{a} < \frac{1}{b}$.

Which of the following is correct?

- A The claim is false in general and the first error is on line II, since ab may be negative or zero.
- B The claim and proof are correct.
- C The first error is on line III.
- D The claim is true but the first error is on line I.
- E The claim is false and the first error is on line III.

- 5 Let f be a function defined for all real numbers. $P: 'f'(a) = 0'$; $Q: 'f$ has a stationary point at $x = a'$. Which one of the following is true?
- A P is necessary and sufficient for Q.
 - B P is necessary but not sufficient for Q.
 - C P is sufficient but not necessary for Q.
 - D P is neither necessary nor sufficient for Q.

- 6** A student argues: 'Every prime number is odd, because if it were even it would be divisible by 2 and so not prime.' Which of the following is correct?
- A** The argument is wrong because 2 is even and prime: being divisible by 2 does not stop 2 itself being prime.
 - B** The argument is correct.
 - C** The argument is wrong because some odd numbers are not prime.
 - D** The argument is wrong because 1 is odd and not prime.
 - E** The argument is wrong because 0 is even.

- 7 For a real number x , $P: 'x > 0'$. $Q: 'x^3 > 0'$. Which one of the following is true?
- A P is necessary and sufficient for Q.
 - B P is necessary but not sufficient for Q.
 - C P is sufficient but not necessary for Q.
 - D P is neither necessary nor sufficient for Q.

- 8** A student attempts to solve the equation $\sqrt{x+3} = x-3$ for real x .
- I** Squaring both sides: $x+3 = (x-3)^2 = x^2 - 6x + 9$.
- II** So $x^2 - 7x + 6 = 0$.
- III** So $(x-1)(x-6) = 0$.
- IV** Hence the solutions are $x = 1$ and $x = 6$.

Which of the following best describes the solution?

- A** The solution is wrong, and the first error occurs on line IV.
- B** The solution is completely correct.
- C** The solution is wrong, and the first error occurs on line I.
- D** The solution is wrong, and the first error occurs on line II.
- E** The solution is wrong, and the first error occurs on line III.

9 Consider the statement: ****for all real x , $2^x > x^2$.**** Which of the following is a counterexample?

A $x = 5$

B $x = 3$

C $x = 0$

D $x = 10$

E $x = 1$

- 10** Events A and B have $P(A) > 0$ and $P(B) > 0$. P : ' A and B are mutually exclusive'. Q : ' A and B are not independent'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

- 11** p and q are real numbers. Which of the following statements is/are true?
- I** If $p^2 < 4q$ then $x^2 + px + q > 0$ for all real x .
 - II** If $x^2 + px + q > 0$ for all real x then $q > 0$.
 - III** If $q < 0$ then $x^2 + px + q = 0$ has two real roots of opposite sign.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 12** Let $f(x) = x^2$ for all real x , and let g be the function whose graph is obtained by translating the graph of $y = f(x)$ by 2 units in the positive x -direction and then reflecting in the y -axis. Which of the following statements is/are true?

I $g(x) = (x + 2)^2$

II $g(x) = (x - 2)^2$

III The graph of g has its vertex at $(-2, 0)$

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

13 p and q are prime numbers with $p + q = 45$. Find pq .

A 247

B 140

C 102

D 86

E 94

F 50

- 14** The product of three consecutive positive integers is N , and the middle one of the three integers is m . Which of the following statements must be true?

- I** N is divisible by 6.
- II** N is divisible by 4.
- III** $N + m$ is a perfect cube.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

15 n is a positive integer greater than 1. Which of the following statements is/are true?

I If n is not prime, then n has a prime factor at most $\sqrt{n}$.

II If n has no prime factor at most $\sqrt{n}$, then n is prime.

III If n is prime, then n has no factor between 1 and n exclusive.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 16** The sums 1 , $1 + 3$, $1 + 3 + 5$, $1 + 3 + 5 + 7$ are 1 , 4 , 9 , 16 . A student conjectures that the sum of the first n odd numbers is n^2 . Which of the following is a valid justification?
- A** Each odd number is one more than an even number, and the even numbers sum to $n(n - 1)$... so the total is n^2 since there are n of them: $n(n - 1) + n = n^2$.
- B** The conjecture is true because n^2 is always odd when n is odd.
- C** The pattern 1 , 4 , 9 , 16 continues with 25 , so the conjecture is true.
- D** The odd numbers form an arithmetic series with first term 1 and difference 2 , so the sum of n terms is $\frac{n}{2}(2 + 2(n - 1)) = n^2$.
- E** Four cases have been checked, which is enough for a pattern.

- 17 The n th term of a sequence is $\frac{n^2 + 1}{2n}$. Which of the following statements is/are true?
- I The 5th term is 2.6.
 - II Every term after the first is greater than the previous term.
 - III Some term of the sequence equals 10.
- A none of them
- B I only
- C II only
- D III only
- E I and II only
- F I and III only
- G II and III only
- H I, II and III

18 The cubic $y = x^3 + ax + b$ has three distinct real roots. Which of the following statements must be true?

I $a < 0$

II $b \neq 0$

III The curve has two turning points.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

19 Let $u_n = n^2 - n + 41$. Which of the following statements is/are true?

I u_n is prime for $n = 1, 2, 3, 4, 5$.

II u_n is prime for all positive integers n .

III u_{41} is divisible by 41.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

20 How many ordered pairs of positive integers (m, n) satisfy $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$?

A 4

B 3

C 2

D 6

E 8

F 5

END OF TEST